

EXPERIMENT: Write a Prolog Program to implement Towers of Hanoi.

Aim

To write and implement a Prolog program for solving the Towers of Hanoi problem using recursion and display the sequence of moves required to transfer all disks from the source peg to the destination peg.

Objectives

- To understand the concept of recursion in Prolog.
- To implement the Towers of Hanoi algorithm using Prolog predicates.
- To find the minimum number of moves required to solve the problem.
- To understand how Prolog uses facts, rules, and logical inference to solve problems.

Algorithm

1. Start the program.
2. Read the number of disks N.
3. If $N = 1$, move the disk directly from the source peg to the destination peg.
4. Otherwise:
 - Move N-1 disks from the source peg to the auxiliary peg.
 - Move the largest disk from the source peg to the destination peg.
 - Move N-1 disks from the auxiliary peg to the destination peg.
5. Continue recursively until all disks are moved.
6. Display each movement.
7. Stop the program.

CODE

```
def towers_of_hanoi(n, source, auxiliary, destination):  
    if n == 1:  
        print("Move disk 1 from", source, "to", destination)
```

```

    return

# Move n-1 disks from source to auxiliary
towers_of_hanoi(n - 1, source, destination, auxiliary)

# Move the largest disk from source to destination
print("Move disk", n, "from", source, "to", destination)

# Move n-1 disks from auxiliary to destination
towers_of_hanoi(n - 1, auxiliary, source, destination)

print("=" * 50)

print("      TOWERS OF HANOI")

print("=" * 50)

n = int(input("Enter the number of disks: "))

if n <= 0:
    print("Please enter a positive number.")
else:
    print("\nSequence of moves:")

    print("-" * 50)

    towers_of_hanoi(n, "Source", "Auxiliary", "Destination")

    total_moves = (2 ** n) - 1

    print("-" * 50)

    print("Total number of moves:", total_moves)

    print("=" * 50)

```

OUTPUT

```
IDLE Shell 3.12.5
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>
= RESTART: C:/Users/shaik arshad basha/AppData/Local/Programs/Python/Python312/x
dc.py
=====
          TOWERS OF HANOI
=====
Enter the number of disks: 2

Sequence of moves:
-----
Move disk 1 from Source to Auxiliary
Move disk 2 from Source to Destination
Move disk 1 from Auxiliary to Destination
-----
Total number of moves: 3
=====
>> |
```

Result

The Towers of Hanoi problem was successfully implemented in Prolog using recursion. For 3 disks, the program successfully generated the required 7 moves.

Conclusion

The program demonstrates how recursion and logical programming in Prolog can be used to solve the Towers of Hanoi problem efficiently. It also shows that the minimum number of moves follows the formula $2^n - 1$.